

\$2.8 million for the whole installation. These figures suggest nearly 90 times the cost of a normal solar rooftop (in China, solar panels for general use cost \$5 per sq.m).

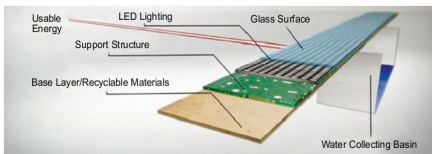
If India decides to pave the foundation of solar roadways, the country will run into similar cost burdens. Therefore the amount of energy generation in return needs to be well-calculated to ensure quick paybacks.

The second concern is about optimal solar capture. The essence of solar panels lies in the amount of solar energy captured on their bed, which is maximised by tilting panels in accordance with the direction of the sun. Solar surfaces on roads will lie as a flatbed. Therefore it needs to be verified whether these will be able to absorb enough energy in this position without the ability to tilt. Additionally, numerous coatings above the solar surface to hold the load of vehicles daily may affect its solar acceptance. If unchecked, this will have a direct impact on the ROI.

Vinesh Kumar, senior manager-projects, SunGarner Energies, says, "There will be major concern about safety and quality of panels. Efficiency depends on the cell quality. To get optimal efficiency in such technology, the cell quality has to be topnotch and protective coatings should also complement the cell."

Another focus area should be the driving safety. Vehicle tyres require a slightly rough surface to balance between acceleration, breaks and directions, aided by the friction between the tyre and the road. In the attempt to make the solar surface absorb maximum sunlight, roughness of the surface might get compromised. So constructors must pay attention to the composition of the coating material.

In addition, infrastructure might prove to be a challenge since arranging such solar structures requires some fine-tuned operation with technical knowhow. Imports may add to the cost since local



Representative image of the constitutive layers of a solar road, where the electronic support structure and the base layer help in collection and distribution of captured energy while other layers act as support elements

solar manufacturing is in developmental stages.

Vyas says, "India does not have the raw material production capacity for photovoltaic cells. This is why the percentage of imports is so high."

Time taken in legal permissions regarding land and other parameters may also add to the list. All these challenges need to be addressed before going ahead with solar road projects.

Way to go

Given all the challenges in solar panel road implementation, there is a lot to be thought out and tested before going ahead with a commercial application. The present cost estimations of such projects direct us towards generic solar rooftops and panel usage as the economical solution for now. However, with more research and mainstream implementation, these materials would become cheaper.

Vyas believes, "In terms of technology and expansion, India is doing

good. Five-seven years down the line, renewable energy utility in India will be extremely improved and technologies like solar panel roads will become feasible."

Increased domestic manufacturing would also add to the advantage. Vyas continues, "China has been providing solar structures at lower cost than India so far. With the government set to impose an import tariff on solar equipment, we will see a substantial decrease in ready-made Chinese panels usage. Indian companies will have to manufacture as well as research domestically. This will change the paradigm of solar implementation and also promote advanced applications like solar roadways."

Vivek Vashista, senior engineer, ABB India, believes solar panel roads will be viable in India. "In fact, our R&D team is already looking into such projects. The return on investment is definitely a key focus area and will be assured," he shares.

Elaborating on ROI, Vinesh adds, "The ROI will depend on the investment amount and volume of the project. The level of energy generation and efficiency of the material will also be major contributors. On an average, the payback period can be expected as eight years."

If successfully implemented, these projects can solve a lot of real-life problems in India today. It is a decision of investing big in view of long-term benefits. **EFY**



India has great scope for leveraging solar energy through its highways